DATA ANALYSIS LEARNING PATH

Statistics

for Data Analysis

Complete notes with formulas, worked examples,
practice questions and Excel · SQL · Python · Power BI code

WHAT'S INSIDE

- Descriptive statistics, probability and distributions
- Sampling, the Central Limit Theorem and confidence intervals
- Hypothesis testing, t-tests, chi-square and ANOVA
- Correlation, regression and A/B testing
- Tool mapping, formula cheat sheet and 12 practice questions

Table of Contents

1. Foundations of Statistics 4

2. Descriptive Statistics 6

3. Data Visualisation for Statistics 9

4. Probability Essentials 10

5. Random Variables and Probability Distributions 11

6. Sampling Distributions and the Central Limit Theorem 13

7. Estimation and Confidence Intervals 14

8. Hypothesis Testing 15

9. Common Statistical Tests 17

10. Correlation and Regression 19

11. Applied Statistics for Analysts 21

12. Statistics in Excel, SQL, Python and Power BI 23

13. One-Page Formula Cheat Sheet 26

14. Practice Questions with Answers 27

15. Glossary and Study Roadmap 28

How to use these notes

Chapters 1–11 build the theory in a sensible order: describe data, understand probability and distributions, then move to estimation, hypothesis testing and regression.

Chapter 12 shows how every concept is done in **Excel, SQL, Python and Power BI**. Chapter 13 is a one-page formula sheet, Chapter 14 has practice questions with answers, and Chapter 15 is a glossary.

Every worked example uses real computed numbers, so you can reproduce them in Excel or Python to check your understanding.

1. Foundations of Statistics

1.1 What is statistics and why analysts need it

Statistics is the science of collecting, organising, analysing, interpreting and presenting data so that we can make decisions under uncertainty. As a data analyst you will use it to answer four kinds of questions:

- **Describe** – what happened? (average sales, spread of delivery times)
- **Explain** – why did it happen? (is the difference between two groups real, or just noise?)
- **Predict** – what will probably happen? (regression, forecasting)
- **Decide** – what should we do? (A/B test results, risk assessment)

1.2 Descriptive vs inferential statistics

	Descriptive statistics	Inferential statistics
Goal	Summarise and describe the data you actually have	Use a sample to draw conclusions about a larger population
Tools	Mean, median, SD, charts, tables, percentiles	Confidence intervals, hypothesis tests, regression, p-values
Certainty	Exact (you computed it from the data)	Probabilistic (always carries uncertainty)
Example	"The average order value in our file is Rs 4,200"	"We are 95% confident the true average order value of all customers is between Rs 4,000 and Rs 4,400"

1.3 Population, sample, parameter and statistic

A **population** is the complete group you care about; a **sample** is the subset you actually measure. A number describing a population is a **parameter**; a number computed from a sample is a **statistic**. Statistics are used to estimate parameters.

Quantity	Population (parameter)	Sample (statistic)
Mean	μ (mu)	$\bar{x}$ (x-bar)
Standard deviation	σ (sigma)	s
Variance	σ^2	s^2
Proportion	p	$\hat{p}$ (p-hat)
Size	N	n
Correlation	ρ (rho)	r

1.4 Types of data and levels of measurement

Level	Type	Properties	Examples	Valid summaries
Nominal	Qualitative	Categories, no order	Gender, city, product category	Mode, counts, percentages
Ordinal	Qualitative	Ordered, gaps not equal	Satisfaction (1–5), education level	Median, mode, percentiles
Interval	Quantitative	Equal gaps, no true zero	Temperature in °C, calendar year	Mean, SD, correlation
Ratio	Quantitative	Equal gaps and true zero	Salary, age, weight, distance	Everything, including ratios ("twice as much")

- **Discrete** numerical data can be counted (number of orders, defects). **Continuous** data can take any value in a range (height, time, revenue).
- **Cross-sectional** data is collected at one point in time; **time-series** data is the same measure tracked over time; **panel** data is both.
- Why it matters: the data type decides which chart, summary statistic and test is allowed. Averaging a nominal code like Product_ID is meaningless.

1.5 Sampling methods

Method	How it works	Note
Simple random	Every member has an equal chance of selection	Gold standard; needs a full list of the population
Systematic	Select every k-th member	Easy; risky if the list has a repeating pattern
Stratified	Split into groups (strata), sample randomly from each	Guarantees representation of subgroups
Cluster	Randomly pick whole groups (e.g. branches), survey everyone in them	Cheap; larger error than stratified
Convenience	Whoever is easy to reach	Biased – avoid for inference
Voluntary response	People choose to respond (online polls)	Biased – extreme opinions over-represented

Common sources of bias

Selection bias (sample not representative), **non-response bias** (those who ignore the survey differ from those who answer), **survivorship bias** (only studying the winners that remain), **measurement bias** (faulty instrument or leading questions). No statistical method can repair a badly collected sample – this is why "garbage in, garbage out" is the first rule of analysis.

1.6 Observational studies vs experiments

In an **observational study** you record what happens without intervening, so you can find association but not prove cause. In a **randomised controlled experiment** you randomly assign subjects to treatment and control groups; randomisation balances out confounders, so a difference can be attributed to the treatment. A **confounding variable** influences both the cause and the effect (ice-cream sales and drownings both rise with temperature).

2. Descriptive Statistics

We will use this small dataset (n = 10, e.g. daily orders handled by an agent) in every example of this chapter:

$$D = \{ 9, 12, 14, 15, 15, 16, 18, 21, 24, 36 \}$$

2.1 Measures of central tendency

Measure	Definition / formula	Example on D	Best when
Mean ($\bar{x}$)	Sum of values divided by n: $\bar{x} = \Sigma x / n$	180 / 10 = **18.0**	Symmetric data without outliers; uses all the data
Median	Middle value of sorted data (average of the two middle values if n is even)	(15 + 16) / 2 = **15.5**	Skewed data or outliers (income, house prices)
Mode	Most frequent value	**15** (appears twice)	Categorical data; finding the most common category
Weighted mean	$\Sigma(w \cdot x) / \Sigma w$	Course GPA, average price weighted by quantity	Values contribute unequally
Trimmed mean	Mean after dropping a % from each tail	Drop 10%: mean of 12…24 = 16.9	Robust compromise between mean and median

> **Key insight**
> The mean is pulled toward outliers; the median is not. Here the single value 36 drags the mean (18.0) above the median (15.5). When mean > median, the data is typically right-skewed.

2.2 Measures of dispersion (spread)

Two datasets can share the same mean and still behave very differently. Dispersion tells you how reliable the average is.

Measure	Formula	Value for D	Notes
Range	max − min	36 − 9 = **27**	Very sensitive to outliers
IQR	Q3 − Q1	20.25 − 14.25 = **6.00**	Spread of the middle 50%; robust
Population variance	$\sigma^2 = \Sigma(x - \mu)^2 / N$	**52.40**	Use only when data is the whole population
Sample variance	$s^2 = \Sigma(x - \bar{x})^2 / (n - 1)$	524 / 9 = **58.22**	Default for samples
Standard deviation	$s = \sqrt{s^2}$	**7.63**	Same units as the data; ~typical distance from the mean
Coefficient of variation	$CV = s / \bar{x} \times 100\%$	7.63 / 18 = **42.4%**	Compares spread across different units or scales
Mean absolute deviation	$\Sigma\lvert x - \bar{x}\rvert / n$	**5.4**	Easy to interpret, less common

Why divide by n − 1 for a sample? (Bessel's correction)

Deviations are measured from the sample mean $\bar{x}$, which is by construction the point closest to the sample values, so dividing by n would systematically *underestimate* the true population variance. Dividing by n − 1 (the **degrees of freedom**) corrects this bias. After $\bar{x}$ is fixed, only n − 1 values are free to vary.

Step-by-step standard deviation

x	$x - \bar{x}$	$(x - \bar{x})^2$
9	-9	81
12	-6	36
14	-4	16
15	-3	9
15	-3	9
16	-2	4
18	+0	0
21	+3	9
24	+6	36
36	+18	324
Σ = 180	**0**	**524**

$$s^2 = 524 / (10 - 1) = 58.22 \qquad s = \sqrt{58.22} = 7.63$$

2.3 Measures of position

- **Percentile**: the value below which a given percentage of the data falls. Quartiles are the 25th, 50th and 75th percentiles (Q1, Q2 = median, Q3).
- **Five-number summary**: minimum, Q1, median, Q3, maximum → for D: 9, 14.25, 15.5, 20.25, 36.
- **Z-score** (standardisation): how many standard deviations a value is from the mean.

$$z = (x - \bar{x}) / s \qquad \text{e.g. for 36: } (36 - 18) / 7.63 = \mathbf{2.36}$$

- z = 0 means exactly average; |z| > 2 is unusual; |z| > 3 is very unusual (a common outlier flag for roughly normal data).
- Z-scores let you compare values from different scales (a mark of 80 in one exam vs 65 in another).

2.4 Detecting outliers

Method	Rule	Result for D
Tukey's IQR rule	Outlier if x < Q1 − 1.5×IQR or x > Q3 + 1.5×IQR	Fences: 5.25 and 29.25 → **36 is an outlier**
Z-score rule	Outlier if |z| > 3 (or 2.5)	z of 36 = 2.36 → flagged only under the stricter 2-SD rule

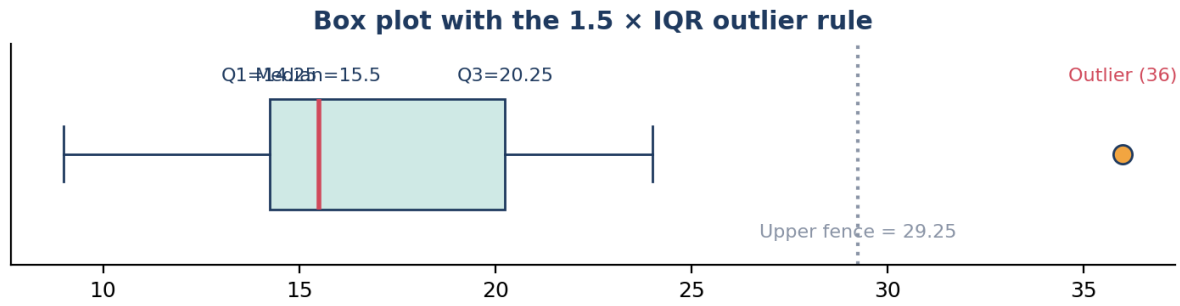


Figure 2.1 – Box plot of D showing quartiles, the upper fence and the outlier.

What to do with an outlier?

Do **not** automatically delete it. First check for a data-entry error, then ask whether it is a genuine rare event. Options: keep it, correct it, remove it (and document why), cap/winsorise it, or switch to robust statistics (median, IQR). Always report what you did.

2.5 Shape: skewness and kurtosis

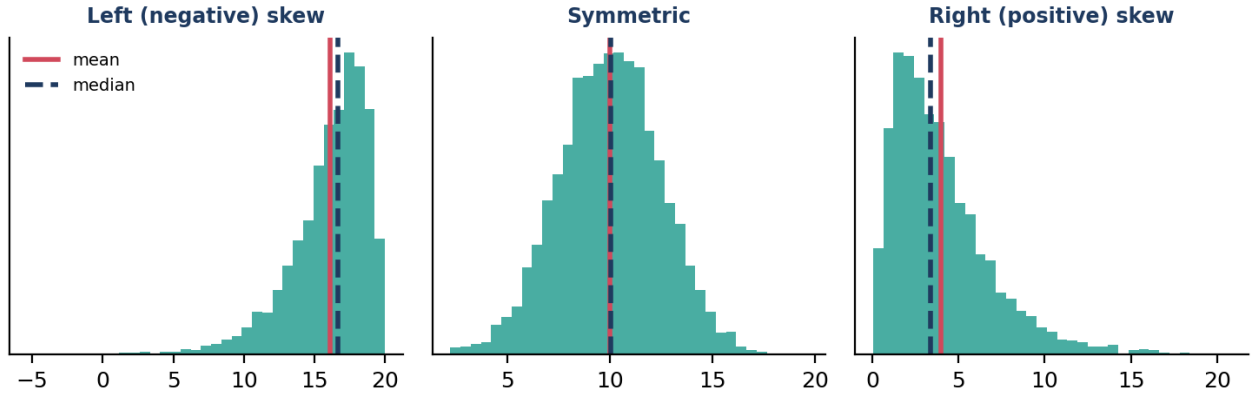


Figure 2.2 – Mean (red) and median (dashed blue) in left-skewed, symmetric and right-skewed distributions.

Shape	Relationship	Typical examples
Right (positive) skew	mean > median > mode; long tail to the right	Income, house prices, order values, response times
Symmetric	mean ≈ median ≈ mode	Heights, measurement errors
Left (negative) skew	mean < median < mode; long tail to the left	Exam scores with a ceiling, age at retirement

- **Skewness** measures asymmetry (0 = symmetric; roughly beyond ±1 is strongly skewed). **Kurtosis** measures tail heaviness: excess kurtosis > 0 means heavier tails (more extreme values) than a normal distribution.
- For right-skewed positive data a **log transformation** often makes the distribution closer to normal.

2.6 Frequency tables

A frequency table counts how often each value or class interval occurs. **Relative frequency** = frequency / n, and **cumulative frequency** is the running total. These feed directly into histograms, Pareto charts and PivotTables.

3. Data Visualisation for Statistics

Question you are asking	Best chart	Why
How is one numeric variable distributed?	Histogram, density plot, box plot	Shows shape, centre, spread, outliers
Compare a numeric variable across groups	Box plot / violin plot, bar chart of means with error bars	Shows medians and spread side by side
Compare categories	Bar chart (sorted)	Length is easy to compare
Trend over time	Line chart	Shows direction and seasonality
Relationship between two numeric variables	Scatter plot (+ trend line)	Reveals linearity, clusters, outliers
Relationship among many variables	Correlation heatmap, scatter matrix	Quickly spots strong pairs
Parts of a whole (few categories)	Stacked bar or a pie/donut with ≤ 5 slices	Pie charts are hard to read with many slices

- **Histogram vs bar chart**: a histogram bins a continuous variable and bars touch; a bar chart compares separate categories and bars have gaps.
- The number of bins changes the story – try several. Rule of thumb: about $\sqrt{n}$ bins, or Sturges' rule $1 + \log_2(n)$.
- Always label axes and units, start bar charts at zero, avoid 3-D effects, and use colour to encode meaning, not decoration.
- Never rely on summary statistics alone: **Anscombe's quartet** is four very different datasets with identical mean, variance and correlation – always plot your data.

4. Probability Essentials

4.1 Basic ideas

- **Experiment**: a process with uncertain outcome. **Sample space (S)**: all possible outcomes. **Event**: a subset of S.
- Probability lies between 0 and 1; the probabilities of all outcomes sum to 1.

$$P(A) = \text{number of favourable outcomes} / \text{total equally likely outcomes}$$

4.2 Rules of probability

Rule	Formula	Meaning
Complement	$P(\text{not } A) = 1 - P(A)$	Often easier to compute "at least one" as 1 − P(none)
Addition (general)	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$	Subtract the overlap so it is not counted twice
Addition (mutually exclusive)	$P(A \text{ or } B) = P(A) + P(B)$	A and B cannot both happen
Multiplication (general)	$P(A \text{ and } B) = P(A) \times P(B \mid A)$	Probability of both happening
Multiplication (independent)	$P(A \text{ and } B) = P(A) \times P(B)$	One event does not affect the other
Conditional probability	$P(A \mid B) = P(A \text{ and } B) / P(B)$	Probability of A once we know B occurred

Mutually exclusive ≠ independent

If two events are mutually exclusive and both have non-zero probability, they are actually **dependent** – knowing one happened tells you the other did not. Independence means $P(A \mid B) = P(A)$.

4.3 Bayes' theorem

$$P(A \mid B) = P(B \mid A) \times P(A) / P(B)$$

Bayes' theorem reverses a conditional probability: it updates a prior belief P(A) after seeing evidence B.

Worked example – medical test

A disease affects 1% of people. A test detects it 95% of the time (sensitivity) and gives a false positive 5% of the time. You test positive. What is the chance you actually have the disease?

P(+ and disease) = 0.01 × 0.95 = 0.0095; P(+ and no disease) = 0.99 × 0.05 = 0.0495

P(disease | +) = 0.0095 / (0.0095 + 0.0495) = **16.1%**. Even with a good test, most positives are false positives because the disease is rare – the **base-rate fallacy**.

4.4 Counting

- **Permutations** (order matters): $nPr = n! / (n - r)!$
- **Combinations** (order does not matter): $nCr = n! / (r!(n - r)!)$ – e.g. 10C3 = 120.

5. Random Variables and Probability Distributions

A **random variable** assigns a number to each outcome of a random process. A **discrete** random variable has countable values and a **probability mass function (PMF)**; a **continuous** one has a **probability density function (PDF)** where probability is the area under the curve, so P(X = exact value) = 0. The **cumulative distribution function (CDF)**, P(X ≤ x), works for both.

$$\text{Expected value (discrete): } E(X) = \Sigma\, x \cdot P(x) \qquad \text{Variance: } \mathrm{Var}(X) = \Sigma\, (x - \mu)^2 \cdot P(x)$$

5.1 Discrete distributions

Bernoulli and Binomial

A **Bernoulli** trial has two outcomes (success with probability p, failure 1 − p). The **Binomial** counts successes in n independent trials with constant p.

$$P(X = k) = nCk \cdot p^k \cdot (1 - p)^{n-k} \qquad \text{Mean} = np \qquad \text{Variance} = np(1 - p)$$

Example: probability of exactly 3 heads in 10 fair coin tosses = 120×0.5^{10} = **0.1172**. Typical uses: number of converted leads out of n calls, defective items in a batch.

Poisson

Counts events in a fixed interval of time or space when events occur independently at a constant average rate λ. Mean = variance = λ.

$$P(X = k) = e^{-\lambda} \cdot \lambda^k / k!$$

Example: a help desk gets 4 tickets per hour on average; P(exactly 2 in an hour) = **0.1465**. Typical uses: arrivals, website hits, errors per page.

5.2 Continuous distributions

Uniform

All values in [a, b] are equally likely: mean = (a + b)/2, variance = $(b - a)^2/12$. Random number generators use it.

Exponential

Models the *waiting time* between Poisson events; right-skewed with mean 1/λ and the memoryless property.

Normal (Gaussian) distribution

The bell-shaped, symmetric distribution defined by mean μ and standard deviation σ. It appears everywhere because of the Central Limit Theorem.

$$f(x) = 1 / (\sigma\sqrt{(2\pi)}) \cdot e^{-(x-\mu)^2 / (2\sigma^2)} \qquad X \sim N(\mu, \sigma^2)$$

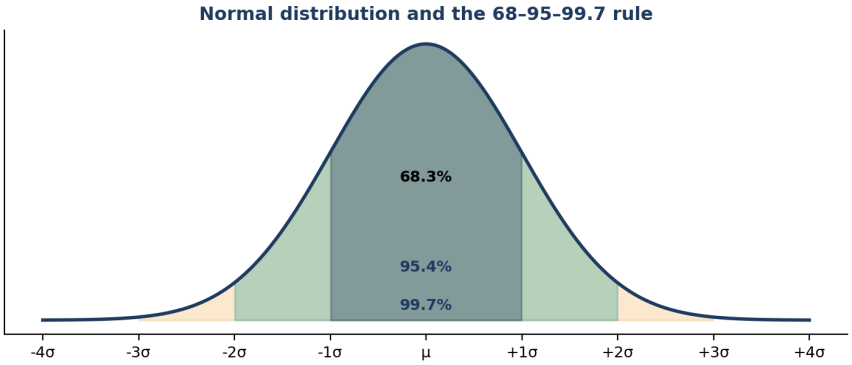


Figure 5.1 – About 68%, 95% and 99.7% of values lie within 1, 2 and 3 standard deviations of the mean.

- **Standard normal** $Z \sim N(0, 1)$. Convert any normal value with $z = (x - \mu)/\sigma$ and then look up the area (z-table, NORM.S.DIST in Excel, scipy.stats.norm).
- Useful z-values: 1.645 (90%), 1.96 (95%), 2.576 (99%) for two-sided intervals.

Worked example – IQ scores N(100, 15)

$P(X < 120)$: $z = (120 - 100)/15 = 1.33 \rightarrow$ area = **0.9088** (about 91% score below 120).

$P(85 < X < 115) = P(-1 < Z < 1)$ = **0.6827** – the 68% rule.

Top 5% cut-off: $x = 100 + 1.645 \times 15$ = **124.7**.

5.3 Distributions used in inference

Distribution	Shape / parameter	Used for
Student's t	Bell-shaped, heavier tails than normal; df = n − 1. Approaches normal as df grows	Tests and CIs for means when σ is unknown (almost always)
Chi-square (χ^2)	Right-skewed, non-negative; df	Tests of independence, goodness of fit, variance tests
F	Right-skewed, non-negative; two df values	ANOVA, comparing variances, overall regression significance

5.4 Summary of common distributions

Distribution	Type	Mean	Excel	Python (scipy.stats)
Binomial(n, p)	Discrete	np	BINOM.DIST(k,n,p,FALSE)	binom.pmf(k, n, p)
Poisson(λ)	Discrete	λ	POISSON.DIST(k,λ,FALSE)	poisson.pmf(k, lam)
Uniform(a, b)	Continuous	(a+b)/2	RAND(), RANDBETWEEN	uniform(a, b−a)
Normal(μ, σ)	Continuous	μ	NORM.DIST(x,μ,σ,TRUE)	norm.cdf(x, mu, sigma)
t (df)	Continuous	0	T.DIST, T.DIST.2T, T.INV.2T	t.cdf(x, df)
Chi-square (df)	Continuous	df	CHISQ.DIST.RT, CHISQ.TEST	chi2.sf(x, df)

5.5 Checking normality

- **Visual**: histogram (bell shape?), Q–Q plot (points on a straight line?).
- **Numeric**: skewness near 0, excess kurtosis near 0.
- **Tests**: Shapiro–Wilk (best for small n), Kolmogorov–Smirnov, Anderson–Darling. The null hypothesis is "data is normal", so a small p-value means *not* normal. With very large n these tests flag trivial deviations, so judge by plots too.

6. Sampling Distributions and the Central Limit Theorem

If you took many random samples of size n and computed the mean of each, those sample means would form their own distribution – the **sampling distribution of the mean**. It is the bridge from descriptive to inferential statistics.

Mean of $\bar{x}$ = μ	Standard error: SE = $\sigma / \sqrt{n}$ (estimated by $s / \sqrt{n}$)

6.1 Central Limit Theorem (CLT)

Statement

For a population with mean μ and finite standard deviation σ, the sampling distribution of $\bar{x}$ becomes approximately **normal** as the sample size grows – **regardless of the population's shape** – with mean μ and standard deviation $\sigma/\sqrt{n}$.

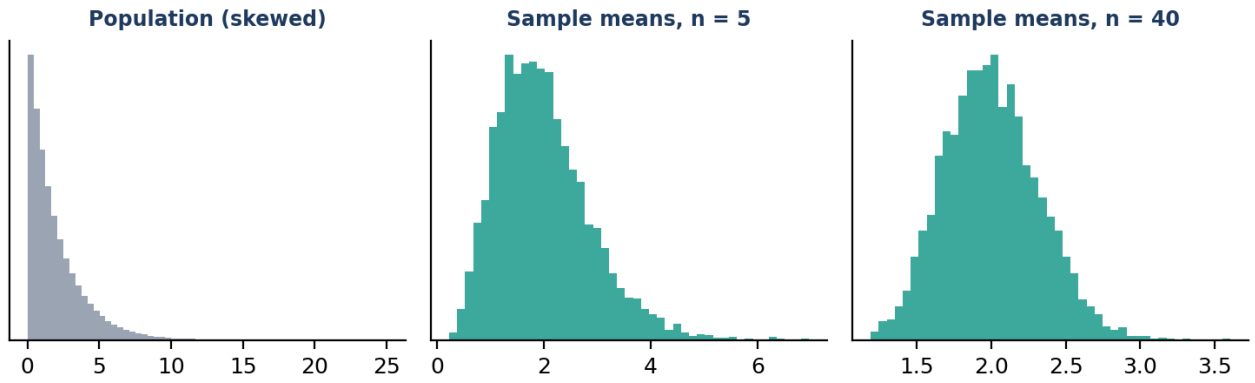


Figure 6.1 – A skewed population (left); means of samples of size 5 and 40 become progressively more normal and narrower.

- Rule of thumb: $n \geq 30$ is usually enough; heavily skewed data may need more, near-normal data needs less.
- **Standard error vs standard deviation**: SD describes the spread of individual values; SE describes the precision of the sample mean. To halve the SE you need four times the sample size.
- **Law of Large Numbers**: as n grows, the sample mean converges to the true mean.
- For a proportion: $SE(\hat{p}) = \sqrt{p(1-p)/n}$. Normal approximation works when $np \geq 10$ and $n(1-p) \geq 10$.

7. Estimation and Confidence Intervals

A **point estimate** (e.g. $\bar{x}$) is a single best guess for a parameter. An **interval estimate** adds a margin of error to express uncertainty.

$$\text{CI} = \text{point estimate} \pm \text{critical value} \times \text{standard error}$$

Parameter	Conditions	Confidence interval
Mean, σ known	Normal data or $n \geq 30$	$\bar{x} \pm z^* \cdot \sigma/\sqrt{n}$
Mean, σ unknown (typical)	Normal data or $n \geq 30$	$\bar{x} \pm t^* \cdot s/\sqrt{n}$, $df = n - 1$
Proportion	$n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$	$\hat{p} \pm z^* \cdot \sqrt{\hat{p}(1-\hat{p})/n}$
Difference of two means	Independent samples	$(\bar{x}_1 - \bar{x}_2) \pm t^* \cdot$ SE of the difference

Confidence level	90%	95%	99%
z^*	1.645	1.960	2.576

7.1 Interpreting a confidence interval

Wrong interpretation

"There is a 95% probability that the true mean lies in this interval." The true mean is a fixed number – it is either in this interval or not.

Correct interpretation

If we repeated the sampling process many times and built an interval each time, about 95% of those intervals would contain the true mean. Practically: "we are 95% confident that the true mean is between a and b."

What makes an interval wider or narrower?

- Higher confidence level → **wider** (more certainty costs precision).
- Larger sample size → **narrower** (SE shrinks with √n).
- Greater variability (larger s) → **wider**.

7.2 Required sample size

$$n = \left(z^* \cdot \sigma / E \right)^2 \text{ for a mean} \qquad n = z^{*2} \cdot p(1-p) / E^2 \text{ for a proportion}$$

Example: σ ≈ 15, desired margin of error E = 3, 95% confidence → $n = (1.96 \times 15 / 3)^2 = 96.04$ → round **up** to **97**. For proportions with no prior estimate use p = 0.5 (the most conservative choice).

8. Hypothesis Testing

Hypothesis testing is a formal procedure for deciding whether sample evidence is strong enough to reject a default assumption about the population.

8.1 The five-step framework

Step	What you do
1. State hypotheses	**Null H_0**: no effect / no difference (contains $=$, $\leq$ or $\geq$). **Alternative H_1**: what you want evidence for ($\neq$, $>$ or $<$).
2. Choose α	Significance level – the Type I error risk you accept (commonly 0.05).
3. Pick the test and check assumptions	Depends on data type, number of groups, sample size, normality (see Chapter 9).
4. Compute the test statistic and p-value	How far is the sample result from what H_0 predicts, measured in standard errors?
5. Decide and interpret	If $p \leq \alpha$, reject H_0; otherwise **fail to reject** it (never "accept" H_0). State the conclusion in plain language.

8.2 The p-value

> **Definition**
> The p-value is the probability of obtaining a result at least as extreme as the one observed, **assuming H_0 is true**. Small p-value → the data would be surprising if H_0 were true → evidence against H_0.

The p-value IS	The p-value is NOT
A measure of how compatible the data is with H_0	The probability that H_0 is true
Smaller when the effect is large or the sample is large	The size or importance of the effect
Compared with a threshold α chosen in advance	A reason to "hunt" until it drops below 0.05 (p-hacking)

8.3 One-tailed vs two-tailed tests

- **Two-tailed** (H_1: $\mu \neq \mu_0$): detects a difference in either direction – the default choice.
- **One-tailed** (H_1: $\mu > \mu_0$ or $\mu < \mu_0$): only when a difference in the other direction is impossible or irrelevant, and decided *before* seeing the data.

8.4 Errors, significance and power

	H_0 is actually true	H_0 is actually false
Reject H_0	**Type I error** (false positive), probability α	Correct – probability = **power** ($1 - \beta$)
Fail to reject H_0	Correct	**Type II error** (false negative), probability β

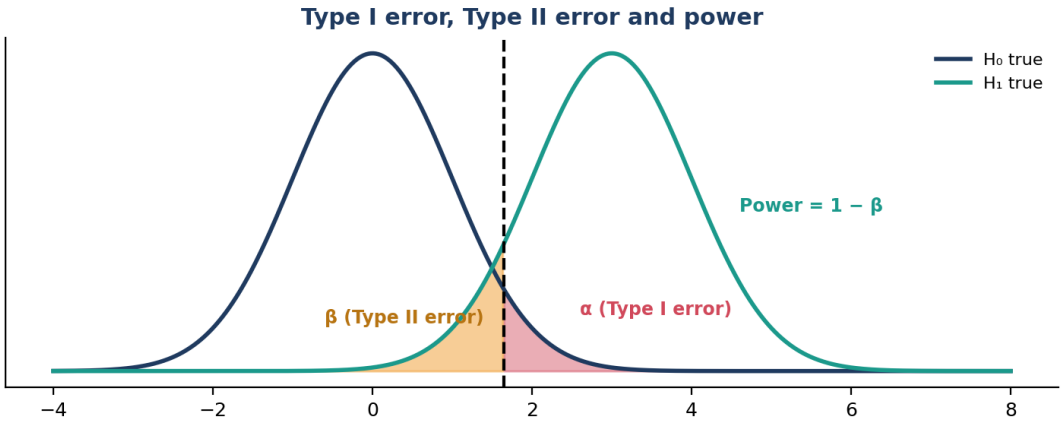


Figure 8.1 – Moving the critical value trades Type I error against Type II error.

- **Power** rises with a larger sample, a larger true effect, a higher α, and lower variability. A typical target is 80%.
- Spam filter analogy: Type I = a good email marked as spam; Type II = spam reaching the inbox. Which error is costlier decides how strict α should be.
- **Statistical vs practical significance**: with huge samples tiny, useless differences become "significant". Always report the **effect size** (e.g. Cohen's d = difference / SD: ~0.2 small, 0.5 medium, 0.8 large) and a confidence interval.

8.5 Worked example – one-sample t-test

Problem

A courier company claims the average delivery time is 30 minutes. A sample of n = 36 deliveries has $\bar{x}$ = 32 minutes and s = 6 minutes. At α = 0.05, is the true mean different from 30?

Step	Work
Hypotheses	H_0: μ = 30 vs H_1: μ ≠ 30 (two-tailed)
Standard error	SE = s/√n = 6/√36 = **1.0**
Test statistic	t = ($\bar{x}$ − μ_0)/SE = (32 − 30)/1.0 = **2.00**, df = 35
p-value	Two-tailed p = **0.0533**; critical value t* = ±2.030
95% CI	32 ± 2.030 × 1 = (**29.97, 34.03**)
Effect size	Cohen's d = 2/6 = **0.33** (small)
Decision	p = 0.053 > 0.05 → **fail to reject H_0**. The evidence is borderline: not enough to say the claim is wrong, but the CI barely contains 30, so collecting more data would be sensible.

Note how the test and the confidence interval agree: 30 lies inside the 95% CI exactly when a two-tailed test at α = 0.05 fails to reject.

9. Common Statistical Tests

9.1 Which test should I use?

Goal	Data	Test	Non-parametric alternative
Compare one mean with a known value	Numeric	One-sample t-test (or z if σ known)	Wilcoxon signed-rank
Compare means of 2 independent groups	Numeric, 2 groups	Independent two-sample t-test (Welch's)	Mann–Whitney U
Compare before/after on the same subjects	Numeric, paired	Paired t-test	Wilcoxon signed-rank
Compare means of 3 or more groups	Numeric, 1 factor	One-way ANOVA (+ Tukey post-hoc)	Kruskal–Wallis
Compare one proportion with a target	Binary	One-proportion z-test	Exact binomial test
Compare proportions of 2 groups	Binary	Two-proportion z-test / chi-square	Fisher's exact test (small counts)
Association between two categorical variables	Counts	Chi-square test of independence	Fisher's exact test
Does data fit an expected distribution?	Counts	Chi-square goodness of fit	–
Linear relationship between two numeric variables	Numeric	Pearson correlation / simple regression	Spearman rank correlation

9.2 t-tests in detail

Test	Test statistic	df
One-sample	$t = (\bar{x} - \mu_0) / (s/\sqrt{n})$	$n - 1$
Independent samples (Welch)	$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$	Welch–Satterthwaite approximation (software computes it)
Paired	$t = \bar{d} / (s_d/\sqrt{n})$, where d = after − before for each pair	$n - 1$ (n = number of pairs)

- **Assumptions**: independent observations; roughly normal data (or n large enough for the CLT); for the classic pooled t-test, equal variances – Welch's version removes this requirement and is the safer default.
- Use a **paired** test when the same person/unit is measured twice (before/after); using an independent test on paired data wastes power.

9.3 Chi-square test of independence

Tests whether two categorical variables are associated. H_0: the variables are independent.

$$\text{Expected} = (\text{row total} \times \text{column total}) / \text{grand total} \qquad \chi^2 = \Sigma\ (\text{Observed} - \text{Expected})^2 / \text{Expected} \qquad \text{df} = (\text{rows} - 1)(\text{columns} - 1)$$

Worked example – ad version vs sign-up

Ad A: 40 of 200 users signed up (20%). Ad B: 60 of 200 signed up (30%). Is sign-up related to the ad version?

Totals: 100 signed up, 300 did not, grand total 400. Expected sign-ups per ad = 200 × 100 / 400 = 50; expected non-sign-ups = 150.

$\chi^2 = (40-50)^2/50 + (160-150)^2/150 + (60-50)^2/50 + (140-150)^2/150 = 2 + 0.667 + 2 + 0.667 =$ **5.333**, df = 1, p = **0.0209**.

$p < 0.05$ → reject H_0: sign-up rate is associated with the ad version. Ad B performs better. Requirement: every expected count should be at least 5.

9.4 One-way ANOVA

ANOVA compares the means of three or more groups in one test. Running many t-tests inflates the false-positive rate (with 3 groups and $\alpha = 0.05$ you get about 14% overall), so ANOVA is used first.

$$F = \text{Variance between groups} / \text{Variance within groups} = MS_{between} / MS_{within}$$

- H_0: all group means are equal. A large F (small p) means at least one mean differs – it does *not* say which.
- Follow up with a **post-hoc** test (Tukey HSD, Bonferroni) to find which pairs differ.
- Assumptions: independence, approximately normal residuals, similar variances across groups (check with Levene's test).

9.5 Proportion tests

$$\text{One proportion: } z = (\hat{p} - p_0) / \sqrt{p_0(1 - p_0)/n}$$

Two proportions use a pooled proportion in the standard error. These are the workhorse of A/B testing conversion rates.

9.6 Assumptions checklist before any test

- Are observations **independent**? (random sampling, no repeated measures unless paired)
- Is the variable's **level of measurement** appropriate?
- Is the data approximately **normal** (or n large)? Else use a non-parametric test.
- Are **variances** similar across groups (or use Welch)?
- Are there extreme **outliers** that distort the result?

10. Correlation and Regression

10.1 Covariance and correlation

$$\text{Covariance: } \operatorname{cov}(x, y) = \Sigma\,(x - \bar{x})(y - \bar{y}) \,/\, (n - 1)$$
$$\text{Pearson } r = \operatorname{cov}(x, y) \,/\, (s_x \cdot s_y)$$

|r|	Rough strength	Sign of r
0.00 – 0.19	Very weak / none	Positive: y tends to rise as x rises
0.20 – 0.39	Weak	Negative: y tends to fall as x rises
0.40 – 0.59	Moderate	r = 0: no *linear* relationship
0.60 – 0.79	Strong	
0.80 – 1.00	Very strong	

- r always lies between −1 and +1 and has no units. $\mathbf{r^2}$ is the proportion of variation in y explained by x.
- **Spearman's ρ** uses ranks – use it for ordinal data, monotonic non-linear relationships, or when outliers are present.
- **Correlation does not imply causation**: possible explanations are x → y, y → x, a third confounding variable, or coincidence (spurious correlation).
- Pearson r only captures *linear* relationships and is sensitive to outliers – always look at the scatter plot.

10.2 Simple linear regression

Regression models how a response variable y changes with a predictor x and lets you predict y.

$$y = \beta_0 + \beta_1 x + \varepsilon \qquad \text{Fitted line: } \hat{y} = b_0 + b_1 x$$

$$b_1 = \Sigma(x - \bar{x})(y - \bar{y}) \,/\, \Sigma(x - \bar{x})^2 = r \cdot (s_y / s_x) \qquad b_0 = \bar{y} - b_1 \bar{x}$$

- **Slope (b_1)**: expected change in y for a one-unit increase in x.
- **Intercept (b_0)**: predicted y when x = 0 (only meaningful if x = 0 makes sense).
- **Least squares** chooses the line that minimises the sum of squared residuals, where residual e = y − ŷ.

Worked example – hours studied vs exam score

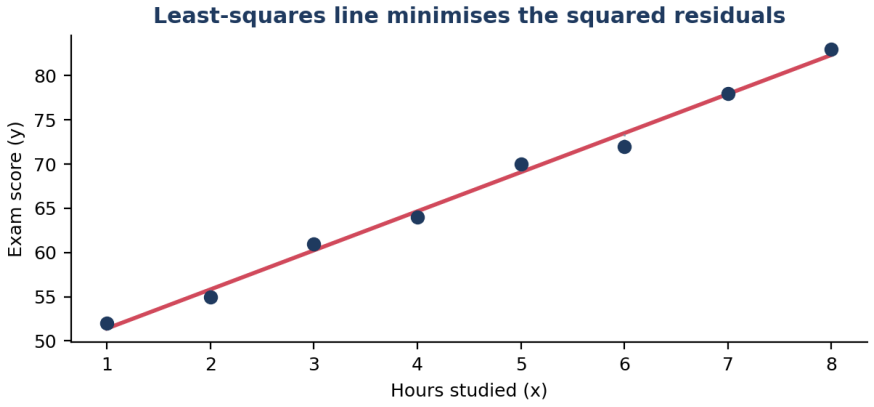


Figure 10.1 – Eight students: scatter plot, fitted line and residuals (dotted).

Result	Value	Interpretation
Slope b_1	4.417	Each extra study hour adds about 4.4 marks
Intercept b_0	47.0	Predicted score with 0 hours
Correlation r	0.9966	Very strong positive linear relationship
R^2	0.9932	About 99.3% of score variation is explained by hours studied
Prediction at x = 5.5	71.3	$\hat{y} = 47 + 4.417 \times 5.5$ (interpolation – safe)

Beware extrapolation

Predicting for x = 20 hours (outside 1–8) would give a score above 100 – the linear pattern cannot hold forever. Only predict inside the range of your data.

10.3 Evaluating a regression model

Metric	Meaning
R^2	Share of variance in y explained by the model (0–1). Never decreases when you add variables.
Adjusted R^2	R^2 penalised for the number of predictors – use it to compare models of different size.
RMSE / MAE	Typical size of prediction error in the units of y. RMSE penalises large errors more.
p-value of a coefficient	Tests H_0: the true coefficient is 0 (variable has no linear effect, given the others).
F-test	Tests whether the model as a whole explains anything (H_0: all slopes are 0).
Confidence interval of a coefficient	Plausible range for the true effect – more informative than p alone.

Assumptions – the LINE mnemonic

- **L**inearity – the relationship between x and y is linear (residual vs fitted plot shows no curve).
- **I**ndependence – residuals are independent (watch out for time-series data).
- **N**ormality – residuals are roughly normal (Q–Q plot); matters mostly for small samples.
- **E**qual variance (homoscedasticity) – residual spread is constant (no funnel shape).

10.4 Multiple regression

$$\hat{y} = b_0 + b_1x_1 + b_2x_2 + \ldots + b_kx_k$$

- Each coefficient is the effect of that variable **holding the others constant**.
- **Categorical predictors** are encoded as dummy (0/1) variables, with one category left out as the baseline.
- **Multicollinearity**: predictors strongly correlated with each other make coefficients unstable. Check the Variance Inflation Factor (VIF); values above 5–10 are a concern.
- **Overfitting**: too many predictors fit noise. Prefer simpler models, use adjusted R^2, and validate on held-out data (train/test split, cross-validation).
- **Logistic regression** is the equivalent for a binary outcome (yes/no); it models the log-odds and outputs probabilities.

11. Applied Statistics for Analysts

11.1 Running an A/B test properly

- **Define** one primary metric (e.g. conversion rate) and the minimum effect worth detecting before starting.
- **Calculate sample size** from the baseline rate, minimum detectable effect, α (0.05) and power (0.80).
- **Randomise** users into A and B; run for full business cycles (at least one full week) and do not stop early the moment p dips below 0.05 ("peeking").
- **Test** with a two-proportion z-test (or chi-square); report the lift with its confidence interval, not just the p-value.
- **Decide** using both statistical significance and business impact.

11.2 Multiple comparisons

If you run 20 independent tests at $\alpha = 0.05$, you expect one false positive by chance alone (probability of at least one $= 1 - 0.95^{20} \approx 64\%$). Correct with **Bonferroni** (use α/m for m tests) or control the false discovery rate (Benjamini–Hochberg).

11.3 Missing data and data quality

- Classify: missing completely at random, at random, or not at random (the missingness itself carries information).
- Options: delete rows (only if few and random), impute with mean/median/mode, model-based or group-wise imputation, or add a "missing" indicator.
- Check duplicates, impossible values (negative age), inconsistent units and date formats before computing anything.

11.4 Classic traps

Trap	What goes wrong
Simpson's paradox	A trend that appears in several groups reverses when the groups are combined (e.g. a treatment looks worse overall but is better in every subgroup because of unequal group sizes).
Base-rate neglect	Ignoring how common something is (see the medical test example).
Regression to the mean	Extreme results tend to be followed by more average ones – not necessarily because of any intervention.
Survivorship bias	Analysing only the cases that survived a filter.
p-hacking / data dredging	Trying many analyses until something is significant.
Ecological fallacy	Assuming group-level patterns hold for each individual.
Average of averages	Averaging group means with unequal sizes without weighting.

11.5 Reporting results in plain language

Template

"Delivery time averaged 32 minutes (SD = 6, n = 36), 2 minutes above the claimed 30 minutes. The difference was not statistically significant at the 5% level (t(35) = 2.00, p = 0.053), and the 95% CI for the true mean was 29.97 to 34.03 minutes. The estimated effect is small (d = 0.33); a larger sample is recommended before drawing a firm conclusion."

Good reports state: the metric, sample size, estimate, uncertainty (CI), test used, p-value, effect size, and the practical meaning.

12. Statistics in Excel, SQL, Python and Power BI

12.1 Excel

Task	Function
Mean, median, mode	AVERAGE, MEDIAN, MODE.SNGL
Spread	STDEV.S (sample), STDEV.P (population), VAR.S, VAR.P, MAX − MIN
Quartiles / percentiles	QUARTILE.INC, PERCENTILE.INC (the .EXC versions use a different method)
Shape	SKEW, KURT
Z-score	STANDARDIZE(x, mean, sd)
Normal / t probabilities	NORM.DIST, NORM.S.DIST, NORM.INV, T.DIST.2T, T.INV.2T
Binomial / Poisson	BINOM.DIST, POISSON.DIST
Confidence interval margin	CONFIDENCE.T(alpha, sd, n), CONFIDENCE.NORM
Hypothesis tests	T.TEST(array1, array2, tails, type), CHISQ.TEST(actual, expected), F.TEST
Correlation and regression	CORREL, RSQ, SLOPE, INTERCEPT, FORECAST.LINEAR, LINEST
Counting / conditional stats	COUNTIFS, SUMIFS, AVERAGEIFS, FREQUENCY

- Enable **Data → Data Analysis (Analysis ToolPak)** for one-click Descriptive Statistics, Histogram, t-tests, ANOVA, Correlation and Regression output tables.
- Use PivotTables for frequency tables and group means; use Insert → Chart for histograms and box-and-whisker charts.

Excel examples

```
=AVERAGE(B2:B101)                    mean
=STDEV.S(B2:B101)                    sample standard deviation
=QUARTILE.INC(B2:B101,3)-QUARTILE.INC(B2:B101,1)   IQR
=(B2-AVERAGE($B$2:$B$101))/STDEV.S($B$2:$B$101)    z-score (copy down)
=T.TEST(B2:B31,C2:C31,2,3)           two-tailed Welch t-test p-value
=CONFIDENCE.T(0.05,STDEV.S(B2:B37),COUNT(B2:B37))  margin of error for a 95% CI
```

12.2 SQL

SQL is excellent for fast descriptive statistics on large tables. Function names differ slightly between databases (PostgreSQL shown; SQL Server uses STDEV / STDEVP and VAR; MySQL uses STDDEV_SAMP).

SQL examples

```
-- Descriptive statistics
SELECT COUNT(*)                                  AS n,
       AVG(amount)                               AS mean,
       STDDEV_SAMP(amount)                       AS sd,
       MIN(amount) AS min_val, MAX(amount) AS max_val,
       PERCENTILE_CONT(0.25) WITHIN GROUP (ORDER BY amount) AS q1,
       PERCENTILE_CONT(0.50) WITHIN GROUP (ORDER BY amount) AS median,
       PERCENTILE_CONT(0.75) WITHIN GROUP (ORDER BY amount) AS q3
FROM orders;

-- Mode
SELECT amount, COUNT(*) AS freq FROM orders
GROUP BY amount ORDER BY freq DESC LIMIT 1;

-- Z-score per row using window functions
SELECT order_id, amount,
       (amount - AVG(amount) OVER ()) / STDDEV_SAMP(amount) OVER () AS z
FROM orders;

-- Group comparison (feeds a t-test / ANOVA done elsewhere)
SELECT region, COUNT(*) AS n, AVG(amount) AS mean, STDDEV_SAMP(amount) AS sd
FROM orders GROUP BY region;

-- Correlation (PostgreSQL / Oracle); regression slope and intercept also available
SELECT CORR(ad_spend, revenue), REGR_SLOPE(revenue, ad_spend), REGR_INTERCEPT(revenue, ad_spend)
FROM campaigns;
```

SQL does not run hypothesis tests natively – the usual workflow is to aggregate in SQL (counts, means, SDs per group) and run the test in Python, Excel or R.

12.3 Python

Python examples (pandas, scipy, statsmodels)

```
import pandas as pd, numpy as np
from scipy import stats
import statsmodels.api as sm

df = pd.read_csv("data.csv")

# --- Descriptive ---
df["amount"].describe()                      # count, mean, std, min, quartiles, max
df["amount"].skew(); df["amount"].kurt()
q1, q3 = df["amount"].quantile([.25, .75]); iqr = q3 - q1
outliers = df[(df.amount < q1 - 1.5*iqr) | (df.amount > q3 + 1.5*iqr)]
df["z"] = stats.zscore(df["amount"], ddof=1)

# --- Normality ---
stats.shapiro(df["amount"])                  # p < 0.05 -> not normal

# --- t-tests ---
stats.ttest_1samp(df["amount"], popmean=30)
stats.ttest_ind(a, b, equal_var=False)       # Welch's t-test
stats.ttest_rel(before, after)               # paired
stats.mannwhitneyu(a, b)                     # non-parametric alternative

# --- ANOVA, chi-square, correlation ---
stats.f_oneway(g1, g2, g3)
chi2, p, dof, expected = stats.chi2_contingency(pd.crosstab(df.ad, df.signed_up))
stats.pearsonr(df.x, df.y); stats.spearmanr(df.x, df.y)

# --- Confidence interval for a mean ---
stats.t.interval(0.95, len(x)-1, loc=x.mean(), scale=stats.sem(x))

# --- Regression ---
X = sm.add_constant(df[["hours", "attendance"]])
model = sm.OLS(df["score"], X).fit()
print(model.summary())                       # coefficients, p-values, R2, F-test, CIs
```

12.4 Power BI (DAX and visuals)

DAX measures

```
Mean      = AVERAGE(Sales[Amount])
Median    = MEDIAN(Sales[Amount])
Std Dev   = STDEV.S(Sales[Amount])
Variance  = VAR.S(Sales[Amount])
Q1        = PERCENTILE.INC(Sales[Amount], 0.25)
Q3        = PERCENTILE.INC(Sales[Amount], 0.75)
IQR       = [Q3] - [Q1]
CV        = DIVIDE([Std Dev], [Mean])

-- Calculated column: z-score of each row
Z Score = DIVIDE( Sales[Amount] - AVERAGE(Sales[Amount]), STDEV.S(Sales[Amount]) )

-- Calculated column: outlier flag (IQR rule)
Is Outlier = IF( Sales[Amount] < [Q1] - 1.5*[IQR] || Sales[Amount] > [Q3] + 1.5*[IQR], 1, 0 )
```

- DAX has no built-in t-test or correlation function. Use a **scatter chart with a trend line** (Analytics pane) for relationships, or a **Python/R visual** to run tests and show p-values.
- The **Analytics pane** adds average, median and percentile reference lines to charts. Use box-plot or histogram visuals (from AppSource or a Python visual) for distributions.
- Use **measures** for statistics that must respond to slicers and filters; use calculated columns for row-level values such as z-scores.

13. One-Page Formula Cheat Sheet

Topic	Formula
Mean	$\bar{x} = \Sigma x / n$
Sample variance / SD	$s^2 = \Sigma(x - \bar{x})^2 / (n - 1)$, $s = \sqrt{s^2}$
Z-score	$z = (x - \mu) / \sigma$
IQR outlier fences	$Q1 - 1.5 \cdot IQR$ and $Q3 + 1.5 \cdot IQR$
Coefficient of variation	$CV = s / \bar{x}$
Conditional probability	$P(A \mid B) = P(A \text{ and } B) / P(B)$
Bayes	$P(A \mid B) = P(B \mid A) \cdot P(A) / P(B)$
Binomial	$P(k) = nCk\ p^k (1 - p)^{n-k}$; mean np; var $np(1 - p)$
Poisson	$P(k) = e^{-\lambda} \lambda^k / k!$; mean = var = λ
Standard error of the mean	$SE = s / \sqrt{n}$
CI for a mean	$\bar{x} \pm t^* \cdot s/\sqrt{n}$
CI for a proportion	$\hat{p} \pm z^* \cdot \sqrt{\hat{p}(1 - \hat{p})/n}$
Sample size for a mean	$n = (z^* \cdot \sigma / E)^2$
One-sample t	$t = (\bar{x} - \mu_0) / (s/\sqrt{n})$, $df = n - 1$
Chi-square	$\chi^2 = \Sigma(O - E)^2/E$; E = row total × col total / grand total
ANOVA	$F = MS_{between} / MS_{within}$
Pearson r	$r = \Sigma(x - \bar{x})(y - \bar{y}) / \sqrt{[\Sigma(x - \bar{x})^2\ \Sigma(y - \bar{y})^2]}$
Regression	$b_1 = r \cdot s_y/s_x$, $b_0 = \bar{y} - b_1\bar{x}$
R^2	$1 - SS_{residual} / SS_{total}$
Cohen's d	$d = (\bar{x}_1 - \bar{x}_2) / s_{pooled}$
Bonferroni	$\alpha_{adjusted} = \alpha / m$

Decision rules to memorise

$p \leq \alpha \rightarrow$ reject H_0. $p > \alpha \rightarrow$ fail to reject H_0. Z-values for 95%: ±1.96. Empirical rule: 68–95–99.7. CLT: $\bar{x} \approx$ normal with SD $\sigma/\sqrt{n}$. Use $n - 1$ for samples.

14. Practice Questions with Answers

Q1. Compute the mean, median, mode and sample standard deviation of 4, 8, 6, 5, 3, 7, 9, 5.

Answer: Mean = 47/8 = **5.875**. Sorted: 3,4,5,5,6,7,8,9 → median = (5 + 6)/2 = **5.5**. Mode = **5**. Sample SD = **2.031**.

Q2. Exam scores are N(70, 10). What proportion of students score above 85?

Answer: $z = (85 - 70)/10 = 1.5 \rightarrow P(Z > 1.5) = 1 - 0.9332 =$ **0.0668** (about 6.7%).

Q3. A sample of 25 customers has mean spend 50 and s = 10. Build a 95% confidence interval for the true mean.

Answer: SE = $10/\sqrt{25} = 2$; $t^*(df = 24) = 2.064 \rightarrow 50 \pm 4.128 =$ **(45.87, 54.13)**.

Q4. A test gives p = 0.03 at α = 0.05. State the decision and interpret it.

Answer: $p < \alpha \rightarrow$ reject H_0. If H_0 were true, results this extreme would occur only about 3% of the time, so there is statistically significant evidence for H_1. This is *not* a 3% chance that H_0 is true.

Q5. Which test compares the average salary of three departments?

Answer: One-way ANOVA (Kruskal–Wallis if the data is clearly non-normal), followed by Tukey post-hoc tests if significant.

Q6. The correlation between ad spend and sales is r = 0.80. What does it tell you?

Answer: Strong positive linear association. $R^2 = 0.64$, so 64% of the variation in sales is linearly associated with ad spend. It does not prove that ads cause sales.

Q7. A call centre receives 3 calls per minute on average. What is the probability of no calls in a minute?

Answer: Poisson with λ = 3: $P(0) = e^{-3} =$ **0.0498**.

Q8. Why is the median preferred over the mean for reporting house prices?

Answer: House prices are right-skewed with a few very expensive properties; these outliers inflate the mean but barely move the median, so the median better represents a typical house.

Q9. What is the difference between a Type I and a Type II error in fraud detection?

Answer: Type I: flagging a genuine transaction as fraud (false alarm, annoys customers). Type II: missing real fraud (financial loss). The costlier error should be controlled more tightly.

Q10. You want a 95% CI with margin of error 3 for a variable with $\sigma \approx 15$. How large must n be?

Answer: $n = (1.96 \times 15/3)^2 = 96.04 \rightarrow$ **97** (always round up).

Q11. Why not simply run several t-tests instead of ANOVA when comparing 4 groups?

Answer: Four groups give 6 pairwise comparisons; each at α = 0.05 inflates the chance of at least one false positive to about 26%. ANOVA tests all means at once with a single controlled α.

Q12. In the regression $\hat{y} = 47 + 4.42x$, interpret the slope and predict y at x = 6.

Answer: Each extra unit of x is associated with a 4.42 unit increase in y. At x = 6: $47 + 4.42 \times 6 =$ **73.5**.

Self-test checklist

- I can explain mean vs median and when each is appropriate.
- I can compute and interpret a standard deviation and a z-score.
- I can state the Central Limit Theorem, explain standard error, and interpret a confidence interval.
- I can write H_0 and H_1, choose a test, and interpret a p-value without the usual mistakes.
- I can describe Type I/II errors and power, and interpret a regression output (slope, R^2, p-value) and its assumptions.

15. Glossary and Study Roadmap

Term	Meaning
Alpha (α)	Significance level; maximum accepted probability of a Type I error
Bias	Systematic error that pushes results in one direction
Confidence interval	Range of plausible values for a parameter, built from a sample
Confounder	Hidden variable related to both cause and effect
Degrees of freedom	Number of values free to vary after constraints are applied
Effect size	Magnitude of a difference or relationship, independent of sample size
Homoscedasticity	Constant variance of residuals across all x values
Null hypothesis (H_0)	Default claim of no effect or no difference
Outlier	Observation unusually far from the rest of the data
p-value	Probability of data at least this extreme if H_0 were true
Power	Probability of detecting a real effect ($1 - \beta$)
Residual	Observed value minus predicted value
Robust statistic	Statistic not strongly affected by outliers (median, IQR)
Standard error	Standard deviation of a sampling distribution
Type I / Type II error	False positive / false negative
Z-score	Number of standard deviations a value is from the mean

Suggested study roadmap

Stage	Focus	Practice in
1	Chapters 1–3: data types, descriptive statistics, charts	Excel: Descriptive Statistics tool, PivotTables, histograms
2	Chapters 4–6: probability, distributions, CLT	Python: simulate coin tosses, sampling distributions with NumPy
3	Chapters 7–9: confidence intervals, hypothesis tests	Excel T.TEST and Python scipy.stats on a real dataset
4	Chapter 10: correlation and regression	Excel LINEST, Python statsmodels; interpret the output
5	Chapter 11: A/B testing and pitfalls	Analyse a conversion-rate dataset end to end
6	Chapter 12: tool mapping	SQL aggregates to feed Python tests; Power BI dashboards with DAX statistics

Tip: choose one real dataset (for example the Superstore sales data) and revisit it at every stage – describe it, test a question about it, and model it. Real data teaches faster than textbook examples.